

# peakSellMTG - Technical Report

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*When has an MTG card peaked, and should you sell it?*

## 1. What the model does

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For a card you own - valued now at its trailing-30-day median price  $P$  - over a holding window  $H \in \{3, 6, 12\}$  months, the model estimates:

- **decline heads**  $P(\text{price falls to } \leq (1-d) \cdot P, \text{ sustained } \geq 14 \text{ days, within } H)$  for the ladder  $d \in \{20\%, 33\%, 50\%, 70\%\}$ ;
- **a no-upside head**  $P(\text{price never sustainably exceeds } 1.10 \cdot P \text{ within } H)$ ;
- the headline  $\text{sell\_score} = P(\text{decline} \geq d) \times P(\text{no upside})$  - high means *peaked, sell now*.

It is the mirror image of the price-spike (buy) model, and it consumes the reprint-risk model's  $p\_reprint_{\{3,6,12\}mo}$  as features - a reprint is the dominant mechanical cause of a price collapse.

## 2. Data & method

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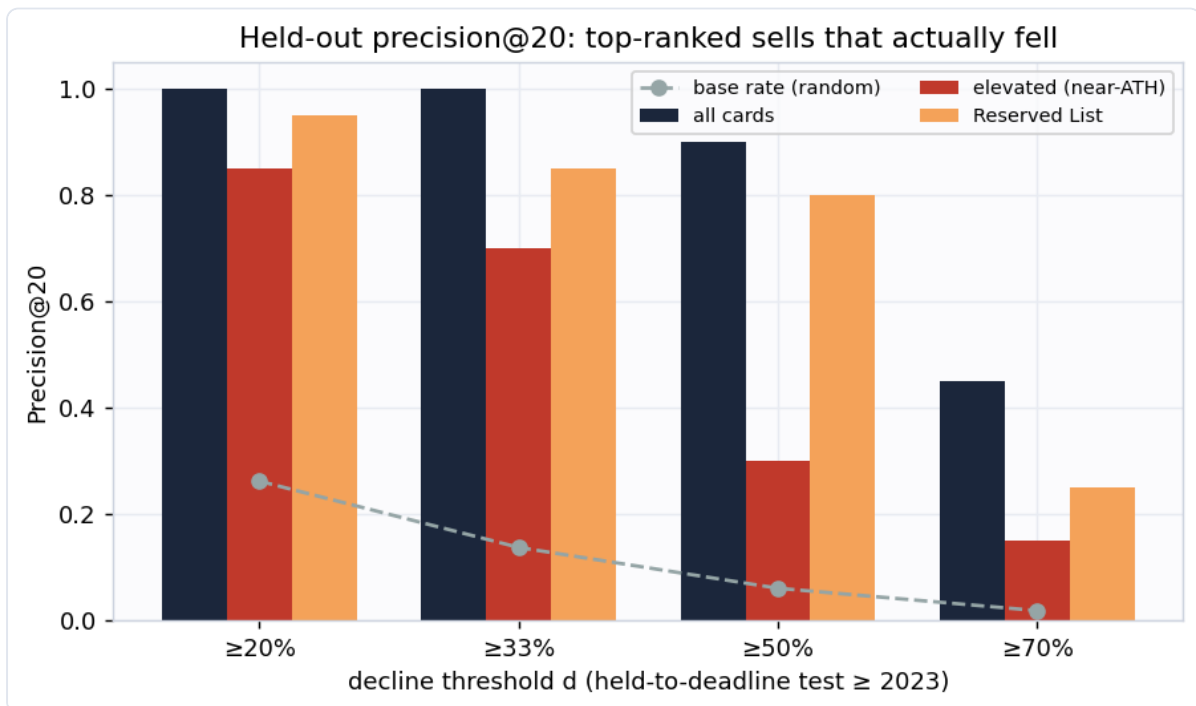
- **Source:** full MTGGoldfish daily price history (171M rows, 30,556 cards, 2010-09  $\rightarrow$  2026-05), Scryfall card intrinsics, EDHREC demand, printings/banlist, and the sibling reprint-risk output. All 571 Reserved-List cards are included.
- **Labels** (point-in-time per  $(\text{card}, \text{month}, H)$ ):  $\text{decline\_mult\_H}$  = forward minimum of the trailing-14-day max ("even your best realistic exit fell this far"), plus  $\text{sustain\_mult\_H}$ .
- **Model:** XGBoost binary heads (`binary: logistic`, PR-AUC objective), holding window  $H$  fed as a feature, isotonic-calibrated; the decline ladder is monotone in  $d$  at predict time.
- **Leakage control:** point-in-time features only; time-blocked split with a per-row embargo  $\geq$  the horizon; an automated audit (`tests/`

`test_no_leakage.py`) incl. a point-in-time check on the all-time-high feature. Train **401,545** / val **205,643** / test **284,078** rows.

- **Elevated slice** (price within 80% of the card's own all-time-high) is calibrated and reported separately - it's the population you actually query.

### 3. Held-out performance (test $\geq$ 2023-01-01)

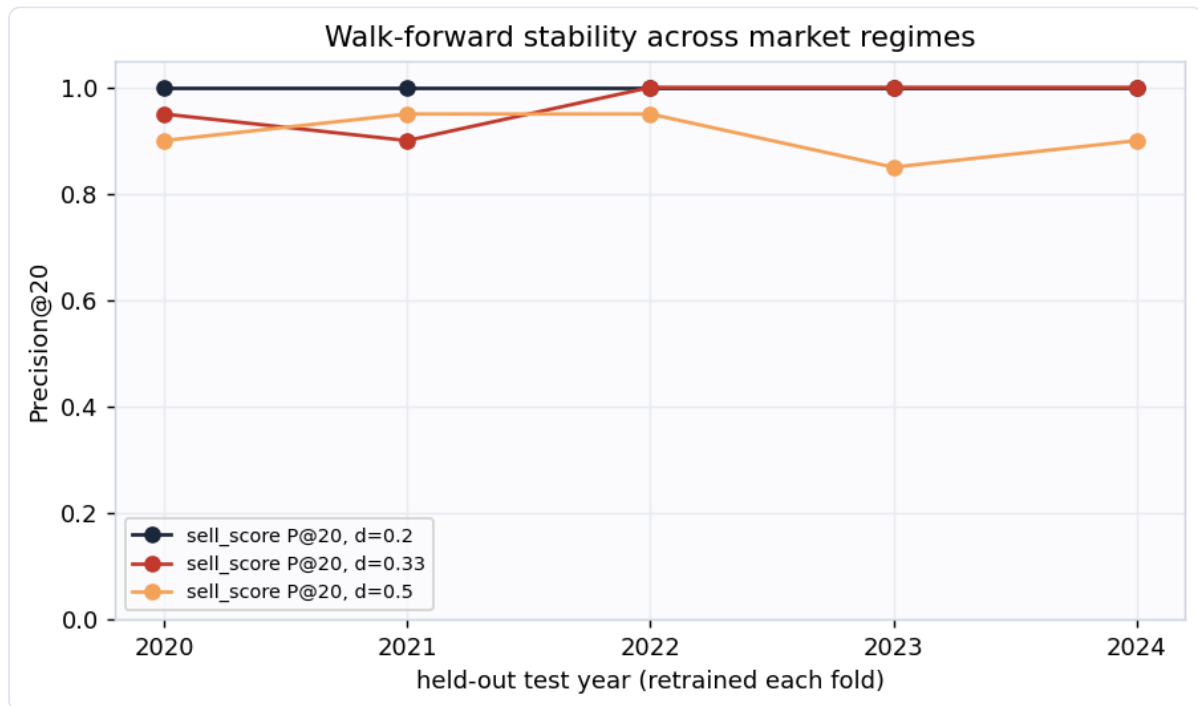
| head         | base rate | PR-AUC | ROC-AUC | P@20 (all) | P@20 (elevated) | P@20 (RL) |
|--------------|-----------|--------|---------|------------|-----------------|-----------|
| decline_0.2  | 26.2%     | 0.613  | 0.808   | 100%       | 85%             | 95%       |
| decline_0.33 | 13.7%     | 0.466  | 0.828   | 100%       | 70%             | 85%       |
| decline_0.5  | 6.1%      | 0.292  | 0.847   | 90%        | 30%             | 80%       |
| decline_0.7  | 1.9%      | 0.091  | 0.829   | 45%        | 15%             | 25%       |
| no_upside    | 66.7%     | 0.855  | 0.770   | 100%       | 100%            | 100%      |



Decline heads reach **ROC-AUC  $\approx$  0.81-0.85**; precision@20 sits far above the random base rate for the actionable thresholds ( $d = 20\text{-}33\%$ ) and is strongest on the **Reserved-List** slice. The  $d = 70\%$  (collapse) head is intentionally weak - those events are rare.

## 4. Stability across regimes (walk-forward)

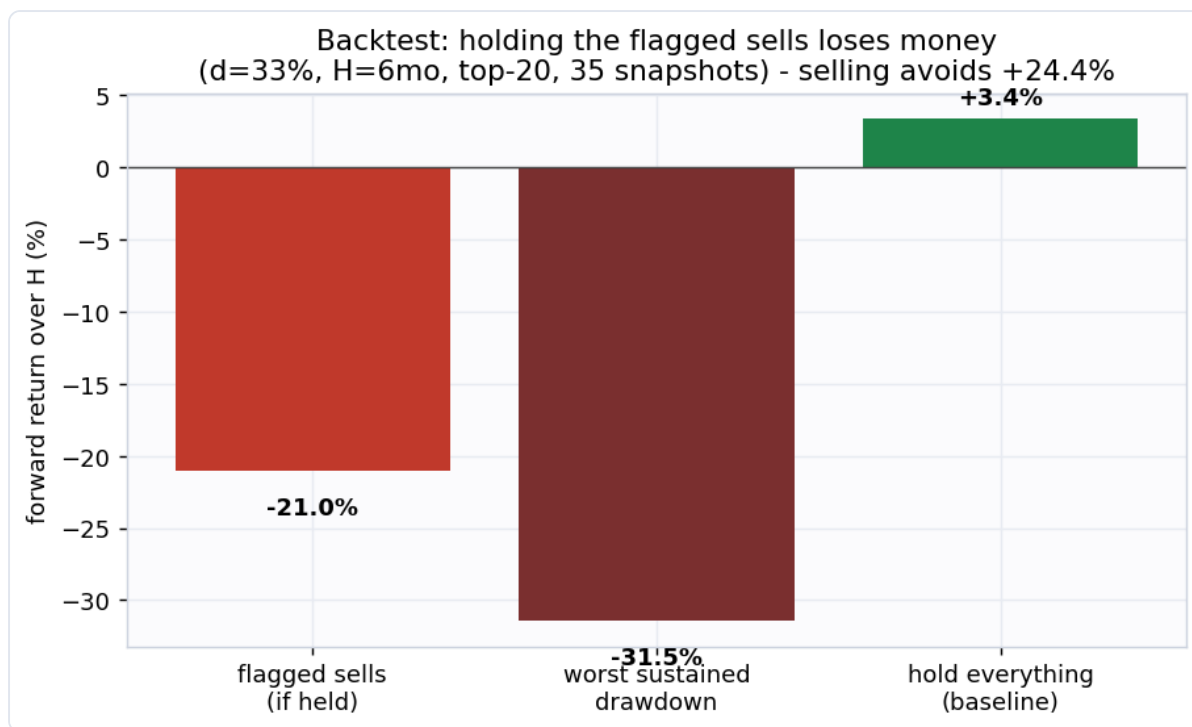
Retrained from scratch on five rolling folds (test years 2020-2024), the `sell_score` top-20 hit rate against realised declines stays in the **85-100%** band every year - not an artifact of one split.



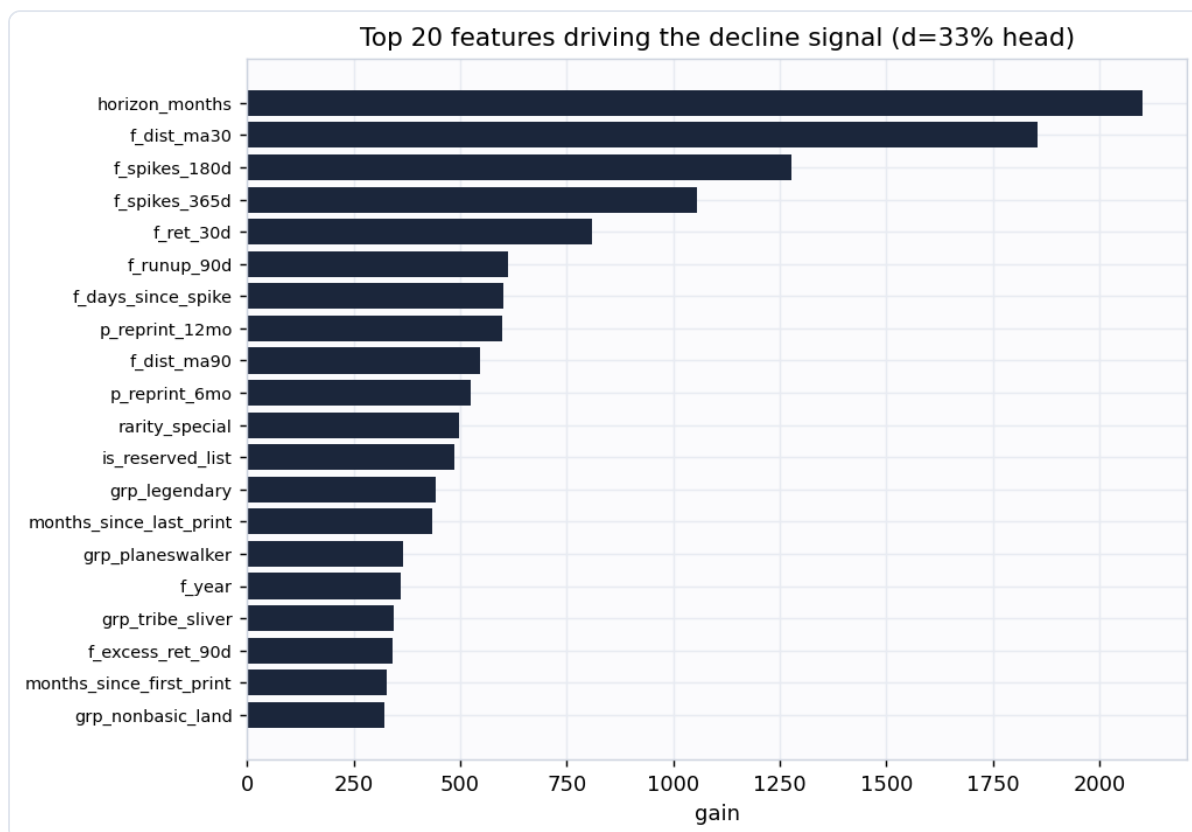
## 5. Does selling actually save money? (backtest)

At each of 35 historical snapshots we "sell" the top-20 flagged cards ( $d = 33\%$ ,  $H = 6\text{mo}$ , elevated only) and measure what *holding* them would have returned:

- flagged sells, if held: **-21.0%**
- holding everything (baseline): **+3.4%**
- **loss avoided by selling: +24.4%**; correct-sell rate 40%



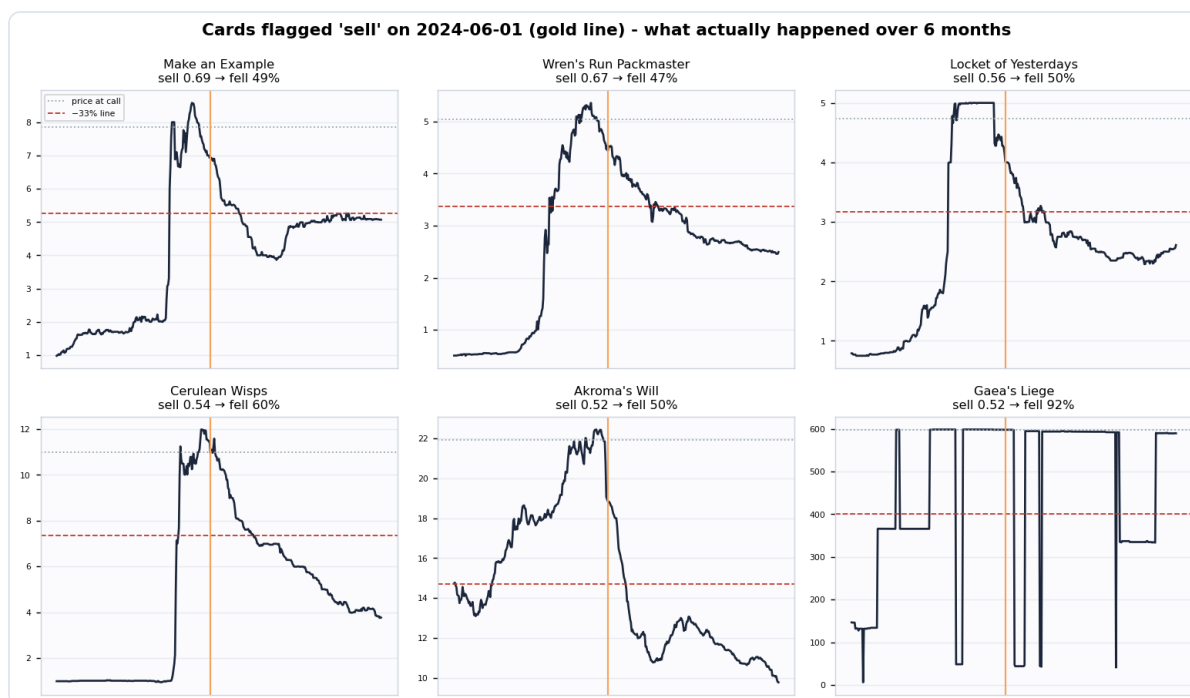
## 6. What drives the signal



Drawdown-from-high, distance-from-moving-averages, RSI, recent run-up, reprint probability, and percent-of-all-time-high dominate - the textbook "overbought / mean-reversion / supply-shock" set.

## 7. Worked examples (out-of-sample)

Cards the model flagged to **sell on 2024-06-01** and what happened over the next 6 months. Of the top-20 flagged that day, **70%** went on to fall  $\geq 33\%$ .



## 8. Caveats

- Trust the **ranking** more than the raw probability. The high-confidence band is  $d = 20-33\%$ .
- Calibration assumes the price regime stays broadly comparable; elevated-slice calibrators target the population you query (owned, appreciated cards), but owned-set performance is an estimate.
- Informational only - not financial advice.

Generated by `scripts/build_report.py` from the trained model + held-out metrics.